Supplementary material to the solution writeup

Appendices A–F

Throughput is accuracy: a verified program-writing agent on a compressed 125B MoE at 1,100 tokens/s on four L4s

Team okiek | ARC Prize 2026, ARC-AGI-2

These appendices accompany our solution writeup. They hold the recipes, hyperparameters and negative results behind its main text: compression (Appendix A), reinforcement learning on the quantization scales (B), the draft head (C), the agent harness (D), scheduling, the deadline and answer selection (E), and the serving engine (F).

Conventions. Scores and differences are percentage points on the 120 public ARC-AGI-2 evaluation tasks, our local set; brackets give intervals. "pass@2 at 100k tokens" means a budget of 100k generated tokens per test input. ×8 and ×79 denote offline pools of 8 and 79 sessions per task, which carry about ±3.5 and ±0.8 points of noise. REPL is the program-writing agent, and FP8 is the unpruned model at FP8.

Appendix A Compression recipe

Calibration. The FP8 model's own attempts on ARC-AGI-2 training tasks: 1,040 sequences on 480 tasks (688 on 320 random tasks at a 64k cap, 352 on 160 harder tasks at 96k), with 48 tasks held out. Teacher-forced replay records routing over 12.6M tokens per layer.

Keep score. Each expert e is scored by

$$\text{score}(e) \;=\; \sum_t g_{t,e}\,\|y_{t,e}\|_2 \;+\; 0.5\,m_e^{\text{call}} \;+\; 0.25\,m_e^{\text{resp}},$$

where $g_{t,e}$ is the gate weight and $y_{t,e}$ the expert's output on token t, and m_e^{call} and m_e^{resp} are its routing mass on tool-call and tool-response tokens. Each of the three terms is normalized.

- $K = 320$ in every layer, including the MTP layer; the top 10 are renormalized over the kept experts.
- A plain usage cut keeps the share of held-out routing weight in Table A.1.
- At $K = 256$, also protecting the experts that carry 90% of header routing raised header coverage from 0.76 to 0.94, with 0 format errors in 90 sessions.
- Every keep set gets at least 30 REPL sessions of format checks.

Table A.1: Held-out routing weight kept by a plain usage cut, by experts kept per layer (K of 512).

Experts kept, K	448	384	320	256	192	128
Routing weight kept	99.3%	97.5%	94.3%	89.4%	82.1%	71.4%

GPTQ.

- Group size 128, asymmetric (fp16 scale, uint8 zero).
- Per-group MSE clip search over 10 ratios, from 1.0 to 0.55; activation ordering; damping mean(diag H)·$(0.01+4/n)$ for n calibration rows.
- Hessians come from router-weighted inputs under the pruned routing. Gate and up are quantized jointly; the down projection's Hessian uses the quantized activations.

Kernels. 4-bit experts run on Marlin. 3-bit experts run on our Triton GEMM (planar layout, `lop3` dequantization), at 374 µs per layer for 64 rows against 476 µs for Marlin 4-bit. The two groups run concurrently: 725.6 MB per layer, 8.7 GB per GPU.

Dense layers.

- Online per-channel FP8 W8A8 for the Gated DeltaNet projections, shared experts and LM head. REPL tests read +5.4 / +4.4 at 40 / 80k tokens (intervals span 0), and it frees memory for 29–31% more KV.
- The other dense weights stay BF16.

Cost against FP8. Table A.2 compares builds by REPL pass@2 at 80k tokens.

Table A.2: Cost of each build against the FP8 model (REPL pass@2 at 80k tokens).

Build	Δ (points)
320 experts, all 4-bit	−5.0
Shipped: 320 experts, 160 at 4-bit and 160 at 3-bit	**−5.9**
384 experts, FP8 (pruning only)	−6.5
512 experts, 3-bit (quantization only)	−6.6
384 experts, 3-bit	−8.3
256 experts, 3-bit	−12.1

Appendix B RL recipe

Parameters. A log-offset t on every per-group scale of the kept experts ($s = s_0\, e^t$, 589.8M), plus the kept router rows (39.3M). Everything else is frozen.

Gradient path.

- A plain-PyTorch model: Gated DeltaNet through fla, sparse attention through flex attention with no-grad block selection, plus hyper-connections and per-layer embeddings. It matches the reference at logit KL 1.5×10^{-4}.
- A custom autograd function rebuilds $W = s\,(q - z)$ from the packed codes, so $\partial L/\partial s = \sum_{\text{group}} \partial L/\partial W \odot (q - z)$.
- About 118 GB peak at 32k tokens.

Objective. REINFORCE with a leave-one-out baseline: $A_i = R_i - \sum_{j \neq i} R_j/(k-1)$ over the task's k samples, with no std scaling. Flat groups, groups smaller than 4 and zero-advantage samples are dropped. For the n sequences of a step, the loss is

$$\mathscr{L} \;=\; -\frac{1}{n \cdot 16{,}384} \sum_{i=1}^{n} A_i \sum_{t} \log \pi(y_{i,t}), \qquad A_i \text{ clipped to } [-2,\, 2].$$

One on-policy pass per round. No KL, PPO clip, entropy bonus, weight decay or gradient clipping.

Table B.1: RL settings.

Setting	Value
Optimizer	Adam, $\beta = (0.9,\ 0.99)$, $\epsilon = 10^{-10}$, fp32 master weights
Learning rates	scales 3×10^{-4}, router 1.5×10^{-5}; 3 warm-up steps, then cosine
Batch and steps	64 sequences per step; 21, 32 and 30 steps in the shipped rounds
Rollouts	shipped harness; $T = 1.0$, top-p 0.95, top-k 20; about 8 per task
Masking	only sampled spans trained; tool output, nudges and forced answers masked; cut at 32,768 tokens
Rollout servers	vLLM with BF16-dequantized copies of the same codes and the current router; trainer–server log-prob gap 0.053 nats per token

Data.

- Only 124 of 948 training tasks gave mixed reward groups: the model already verified 99.0% of sessions.
- So we added leave-one-pair-out variants, in which one demonstration becomes the test.

- Later rounds added ARC-AGI-1 tasks and RE-ARC-derived tasks: fresh instances, the hardest decile, and rule compositions.

Rewards. The first round-1 reward led to abstention; round 1 shipped with the corrected one (Table B.2).

Table B.2: Reward per session outcome.

Outcome	First round-1 reward	Corrected
Verified, right	+1	+1
Verified, wrong	−1	−0.6
Unverified, right	+0.5	+0.5
Unverified, wrong	−0.2	−0.6

Rounds. Table B.3 gives each round's gain, as Δ pass@2 at 100k tokens, ×8, against a control sampled at the same time. Rounds 3 and 2 tie at ×79: −0.18 [−0.99, +0.64]. Rejected: augmentation (−3.7), PPO clip (−0.4), trainable norms (+0.1), credit to 64k tokens (−1.9), and relabelling with fresh RE-ARC instances (−1.7).

Table B.3: The shipped RL rounds.

Round	What changed	Δ pass@2
1	Corrected reward (1,346 samples)	**+7.0 [+3.8, +10.6]**
2	Positive rewards × max(0.2, 1 − tokens/98,304), only on groups that disagree on correctness	+1.7 [−0.8, +4.2]
3	$A = +1$ for each task's 2 shortest right samples (903 samples)	+2.1 [−1.1, +5.4]

Compute and hygiene.

- October 28–30, on up to 24 H200 GPUs.
- Evaluation ids are asserted out, and external pools are filtered by a grid hash that ignores symmetries and re-colouring.
- One round-1 training task shared a concept with a public evaluation task (≤0.8 points locally); it was removed from later rounds.

Appendix C Draft head

Objective. At draft step j, let $\alpha_j = \sum_v \min\left(p_j(v), q_j(v)\right)$ over the target's top 20 tokens v, where p is the target model and q the draft head. The expected accepted length is

$$E = \sum_s \prod_{j \le s} \alpha_j,$$

and the loss is $-E$+0.3 KL, with the draft steps unrolled in training, FP8 KV emulated, and prose positions weighted ×2.

Data. 3,328 RL rollouts (99M tokens), with hidden states and top-20 log-probs dumped from the deployed low-bit stack; 1,200 sequences trained, 26 tasks held out.

Training. 100.7M trainable parameters (dense MTP tensors, router rows, scale offsets); AdamW at 2×10^{-5} for dense and router, 3×10^{-4} for scales; 300 updates of 4 sequences; 40,960-token crops; gradient clip 1.0; 1.5 h on one GPU.

Results. Closed loop: 2.79 → 3.13 tokens per step. On Kaggle: 3.1–3.3 tokens per step.

Drafting choices.

- A 4th draft adds 9% acceptance but 8% step time: −2.4% net.
- Sampled drafts pay off only with the hot-vocabulary head: 620 vs 519 tokens/s.

- Block verification +0.9%; top-20 draft distributions +2.2%.
- Prose accepts 2.48 tokens per step, code and grids about 3.3.

Appendix D The agent harness

Prompt. Each grid is shown under `Input (HxW):` with one row of digits per line. The instructions ask for a general `def transform(grid)`, the full program in every call, and one fenced `json` answer per test input, followed by the `ak` API sheet.

Decoding. The highest reasoning effort (medium scored 2–4 lower); $T = 1.0$, top-p 0.95, top-k 20 ($T = 0.7$ cost 1.9).

Sandbox. `python -I` as an unprivileged user: 2 GiB, 21 s of CPU, a 25 s wall clock, and an audit hook that blocks processes, sockets and outside writes. Up to 32 run in parallel.

Session loop.

- Results are appended to the raw context.
- After 49,152 tokens of thinking without a tool call, a nudge to test the best hypothesis in code.
- Imports and needed definitions carry over between cells.

The `ak` library. describe, check (with leave-one-out and equivariance), objects, diffs, D8, scaling and tiling, sub-grid search, flood fill, lines and gravity, periodicity, symmetry completion, 16 size rules, colour maps, invariants.

Invariants. Palette subsets, output ≠ input, at least 2 colours, colours shared by all outputs, the same colour count, counts that never rise or never fall, kept non-background cells, background-only changes. Calibrated on training tasks, with ≤1.1% false alarms.

Feedback. stdout up to 6,000 characters, the traceback tail, and `transform() check: n/m` with a masked diff of the first two failing examples (correct cells shown as `.`).

Failure atlas (960 FP8 sessions).

- 130 of 230 wrong outputs were near misses.
- Wrong verified programs most often met a test feature unseen in training (23 of 64).
- The first `transform` came at 24k tokens in verified sessions, 53k in failures.

Negative within-sample tests. Every change in Table D.1 lost points.

Table D.1: Harness changes that tested negative within sample.

Change	Δ
Review turn	−0.5 [−1.1, −0.1]
Alternative programs	−1.0
Extra feedback	−0.1
Generalization prompt	−0.2
Nudge toward code	−1.0
Early abort at 30 / 40 / 50 / 60k tokens	−18.4 / −11.2 / −6.4 / −2.9

Appendix E Scheduling, deadline and selection

Scheduling. At most 2 concurrent sessions per task after its first. Settle-last: +0.82 [+0.20, +1.44] in replay. Verified-first −3.3, unverified-first −1.1, token-fair −1.7.

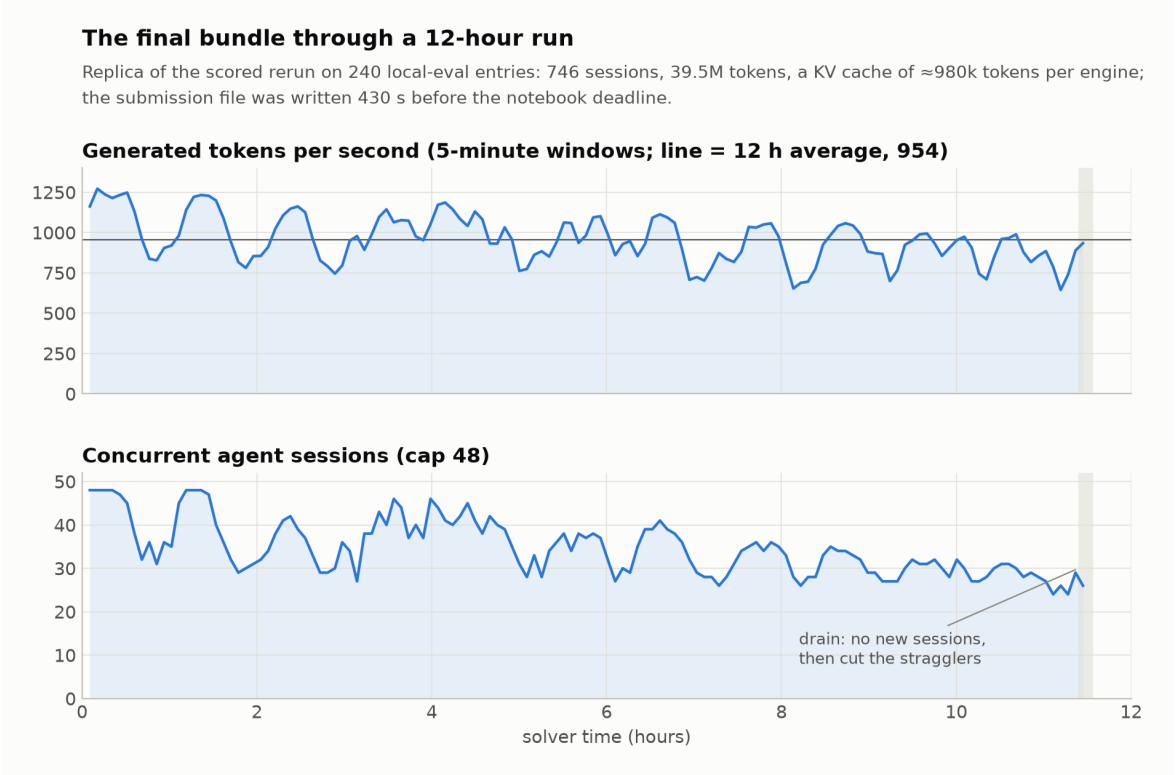


Figure E.1: The final bundle through a 12-hour run: tokens/s and concurrent sessions.

Caps.

- 98,304 generated tokens within a 122,880-token context.
- 48k and 32k caps cost 4–5 and 14 points against 80k.
- 262k caps added +3.75 [−2.5, +9.4] for 11% more tokens. Sessions ending above 100k of context are right 15–16% of the time but use 39–44% of tokens.
- Tail sessions, sized to the time left (at least 16,384 tokens), produce 17% of tokens.

Admission. A new session starts only while

$$\sum_i \max(c_i,\ \tau) + \tau \ \le\ C_{\mathrm{KV}}, \qquad \tau = \max\left(36\mathrm{k},\ 0.6\,\bar{c}_{50}\right),$$

where c_i is the context of running session i, C_{KV} the KV capacity, and $\bar{c}_{50}$ the mean final context of the last 50 sessions. Going to 48 sessions added 3–7% tokens.

Deadline.

- The cut comes 3–10 min before the deadline, set from the measured generation rate. Sessions then stop at their next 2,048-token chunk, run any open call and answer.
- The submission is rewritten atomically after every session. An input-echo file exists from the first minute, and a final cell validates the file and writes an emergency copy.

Robustness.

- A length guard ended device asserts that had crashed 5 of 8 full runs.
- Capping the recurrent-state fold made session resumes exact.
- Reading files in consumption order, plus a compile cache, cut boot from 592 s to 245 s.
- Supervised server restarts and session retries.

Selection.

- Programs verified before 30k tokens have precision 0.80; after 60k, 0.48.
- Ranking: votes, the symmetry and test-only-code checks, branch count, then session order; then forced answers and partial programs.
- `attempt_2` fill: +0.22.
- Dropped: diverse `attempt_2` (−0.10 to −0.48), two-build ensembles (no better than averaging).

Appendix F Engine details

Layout.

- Two TP2 engines with 4-way expert parallelism: total KV 1.71M tokens, against 1.17M with TP4.
- Full-decode CUDA graphs up to 128 tokens: +5.6% tokens/s, +10% KV.
- Prefix caching instead of sessions hit only 38% of the time and cost 11% KV.

Collectives.

- The tensor-parallel all-reduce is one Triton kernel over `cudaHostRegistered` shared memory, with system-scope flags, fp32 accumulation and 16 thread blocks.
- Gathers and scatters use 32 MB host slots; the logits gather goes through host memory too.
- Decode step: 133 → 117 ms.

int8.

- Dispatch at 0.516× the bytes.
- Reduce-scatter peer shares in int8 per 128-element group, with fp32 scales; the local share stays BF16; only above 24 tokens per rank.
- KL $1.34–1.37 \times 10^{-2}$, against a floor of 1.27×10^{-2}. An int8 all-reduce gained nothing.

Kernels. Together −3.7% per step and +2.6% KV:

- Router before the shared expert, about 0.25 ms per step.
- Concurrent 4/3-bit groups with fused cast-and-add, −36 to −51 µs per layer.
- L4-tuned Marlin tiles, +2.0%.
- Overlapped Gated DeltaNet projections, about 0.2 ms per step.

Recurrent state. The fold is deferred into the next verify step: −34% update time.

Memory.

- The n-gram table is 320M rows × 160 FP8 values in one shared-memory file; each rank loads a quarter, and a Triton gather runs on a side stream. The 1.18 GiB token embedding is shared the same way.
- FP8 KV for the MTP layer too (BF16 there cost 4.8–7%).
- Recurrent-state groups 6 → 4: +6.3% tokens/s.
- Shared top-k: +2.7% KV.
- MoE chunk 2,176: +2.8% KV.
- Allocator and NCCL memory settings: +2–6% KV.

Other. Batch-sharded sampling: +2.6%. RC8's flags over RC6's: +7.0%.

Dropped. Memory utilization 0.94 (out of memory), placing experts by routed bytes (+0.4%), batch-token caps (−1.9 to −15%), and five custom kernels that were 1.16–1.46× slower.